MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, BCD/BINARY, DIGITAL, KEY CODE ZEZ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic BCD/binary converter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.151 failures per million operating hours for class I or 0.151 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZEZ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/272-1 for class I and M28787/272-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for V_{OUT} (low-loaded) apply only for a power supply voltage of 7 V dc +0, -1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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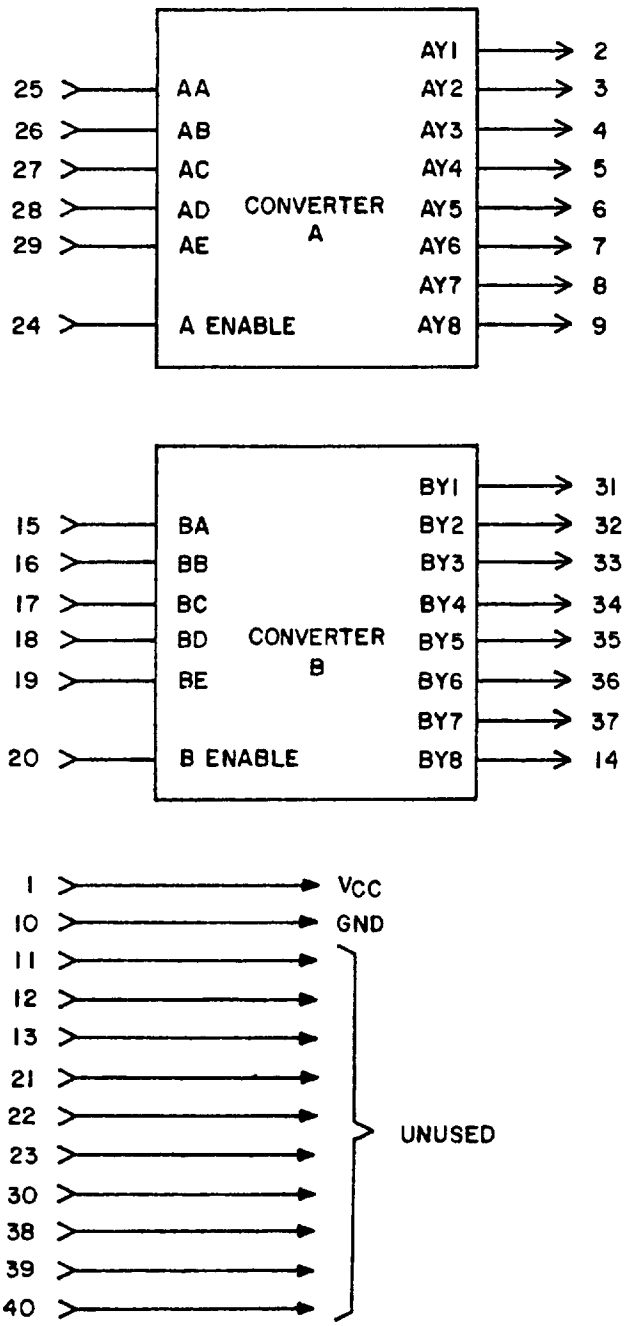


FIGURE 1. ZEZ module function diagram, BCD/binary converter.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
A2	Unused pin isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B16	V_{OUT} (low loaded)	IV	S	0	1	0	1	0	1	0	1	0	1	V
C1-C12	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D12	Input clamp diode voltage	VI	X	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	-0.2	-1.5	V
E1-E34	Functional	VII	X	← See table VII, no failure →										
F1-F16	Output voltage, low level	VIII	X	0	400	0	400	0	400	0	400	0	400	mV
G1-G16	Output leakage current, high level	IX	X	0	100	0	100	0	100	0	100	0	100	µA
H1-H12	Input current, high level	X	X	0	40	0	40	0	40	0	40	0	40	µA
J1-J12	Input current, low level	XI	X	0.00	-1.08	0.00	-1.08	0.00	-1.08	0.00	-1.08	0.00	-1.08	mA
K	Power supply current	XII	X	2	198	2	198	2	198	2	198	2	198	mA
L1-L16	Transition time (t_{THL})	XIII	S	5	30	5	25	5	35	5	25	5	40	ns
M1,M2	Galloping read access pattern	XIV	Q	← See table XIV, no failure →										
N1-N16	Enable recovery time	XV	S	5	55	5	50	5	60	5	45	5	63	ns
P	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
Q	Power supply filter capacitance	XVII	S	3.2	6.2	3.2	6.2	3.2	6.2	3.2	6.2	3.2	6.2	µF

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R1-R12	Input capacitance	XVIII	Q	2	15	2	15	2	15	2	15	2	15	pF
S1-S12	Output capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through S shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through S in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Q shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Q, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests D through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through S shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Supply voltage	4.5 to 5.5	4.5 to 5.5	V dc
Maximum power dissipation (5.5 V operation)	1.089 (outputs not loaded)	1.089 (outputs not loaded)	W
Input voltage			
Low level	-0.5 minimum, 0.8 maximum	-0.5 minimum, 0.8 maximum	V
High level	2 minimum, V_{CC} maximum	2 minimum, V_{CC} maximum	V

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	Class I	Class II	Unit
Input current			
Low level (V_{IL} = 0 V)	-1.08 maximum	-1.08 maximum	mA
(V_{IL} = 0.4 V)	-1 maximum	-1 maximum	mA
High level (V_{IH} = 2.4 V)	40 maximum	40 maximum	µA
(V_{IH} = 5.5 V)	1 maximum	1 maximum	mA
Output voltage			
Low level (I_{OL} = 12 mA)	400 maximum	400 maximum	mV
Output drive current			
Low level (V_{OL} = 0.4 V)	12 maximum	12 maximum	mA
Output leakage current			
High level (V_{OH} = 5.5 V)	100 maximum	100 maximum	µA
Capacitance			
Input	15 maximum	15 maximum	pF
Output	15 maximum	15 maximum	pF

* 6.5 Module function. This module contains two independent 6 bit BCD to binary converters. These converters may be interconnected to form converters of greater length. The module also performs BCD to 9's complement and BCD to 10's complement conversion.

6.5.1 Input to output relationship. Each digital converter, BCD/binary, module has the input to output relationship of table II input conditions other than those shown produce high levels on all outputs. There is a separate enable on each converter, a low level on the enable input allows the converter to function, a high level on the enable input produces a high level on all outputs regardless of the condition of the other inputs. Output Y6, Y7, and Y8 are not used in BCD to binary conversion; and outputs, Y1, Y2, Y3, Y4, and Y5 are not used in BCD to 9's or 10's complement. Module timing is defined on figure 2.

6.5.2 Parallel inputs and outputs. Paralleling of inputs and outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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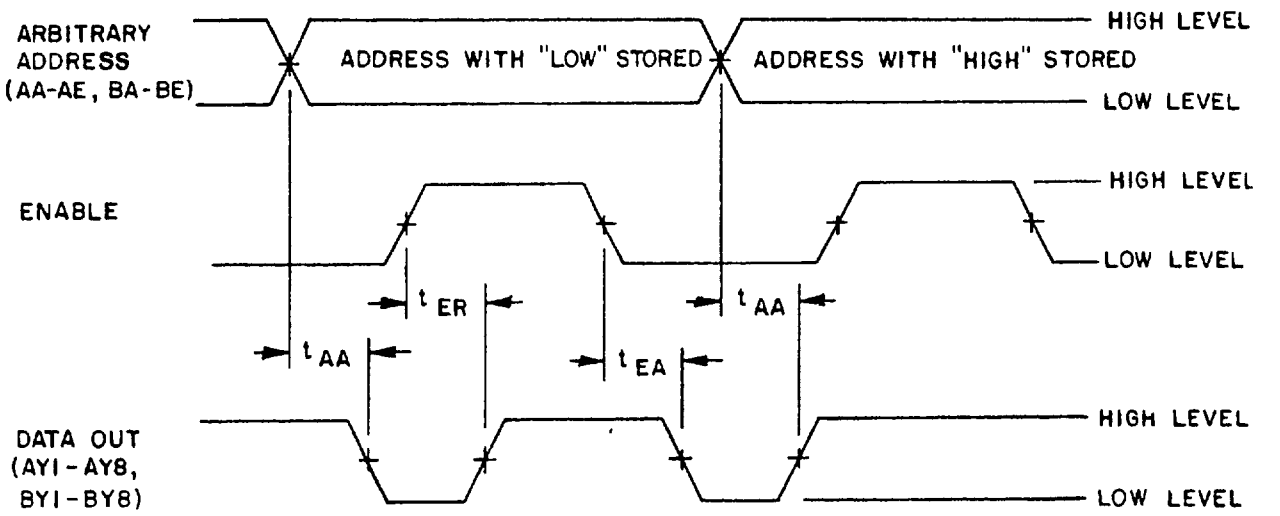


FIGURE 2. Dynamic characteristic waveforms.

TABLE II. Dynamic characteristics and function table.

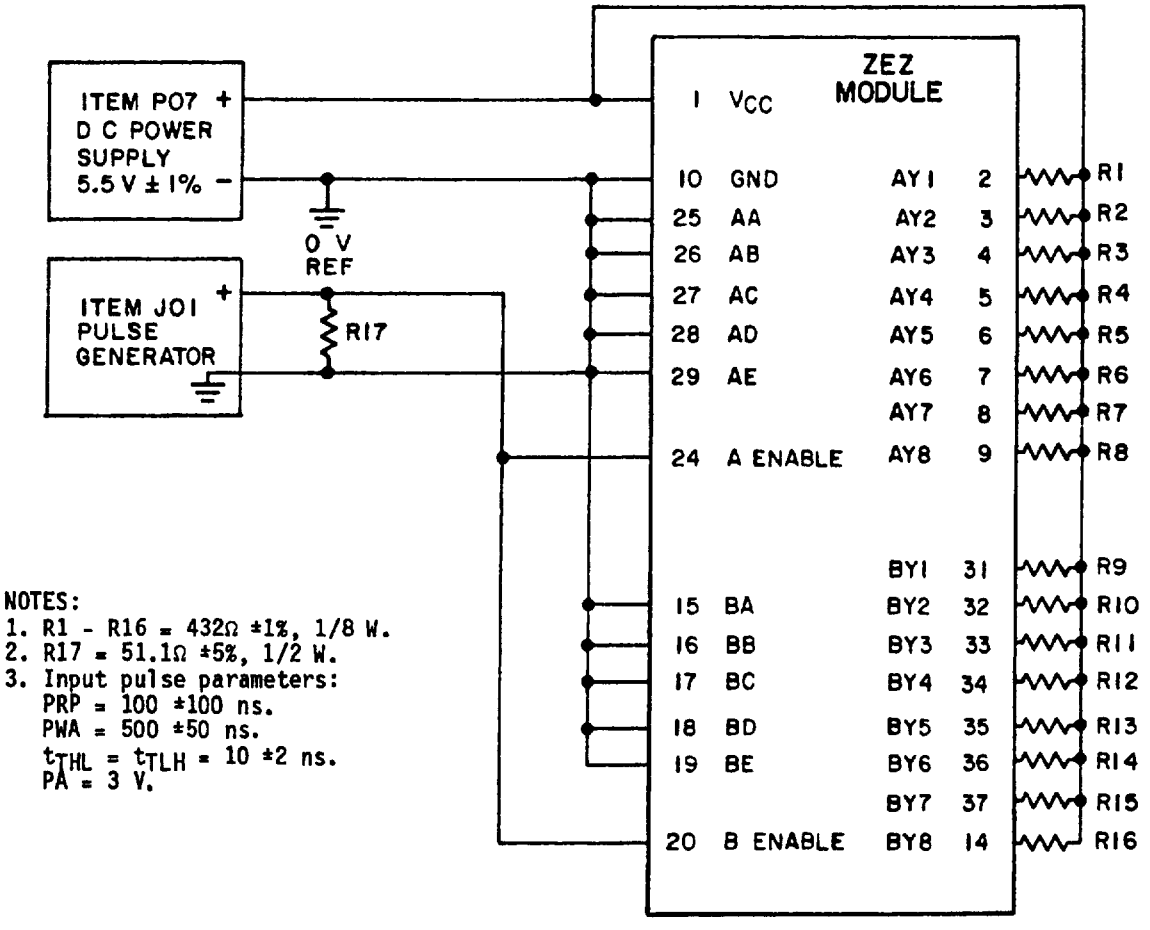
BCD to binary conversion

BCD words	Inputs E	D	C	B	A	Enable	Outputs Y5	Y4	Y3	Y2	Y1
0-1	L	L	L	L	L	L	L	L	L	L	L
2-3	L	L	L	L	H	"	L	L	L	L	H
4-5	L	L	L	H	L	"	L	L	L	H	L
6-7	L	L	L	H	H	"	L	L	L	H	H
8-9	L	L	H	L	L	"	L	L	H	L	H
10-11	L	H	L	L	L	"	L	L	H	L	H
12-13	L	H	L	L	H	"	L	L	H	H	L
14-15	L	H	L	H	L	"	L	L	H	H	H
16-17	L	H	L	H	H	"	L	H	L	L	L
18-19	L	H	H	L	L	"	L	H	L	L	H
20-21	H	L	L	L	L	"	L	H	L	H	L
22-23	H	L	L	L	H	"	L	H	L	H	H
24-25	H	L	L	H	L	"	L	H	H	L	L
26-27	H	L	L	H	H	"	L	H	H	L	H
28-29	H	L	H	L	L	"	L	H	H	H	L
30-31	H	H	L	L	L	"	L	H	H	H	H
32-33	H	H	L	L	H	"	H	L	L	L	L
34-35	H	H	L	H	L	"	H	L	L	L	H
36-37	H	H	L	H	H	"	H	L	L	H	L
38-39	H	H	H	L	L	"	H	L	L	H	H
Any	X	X	X	X	X	H	H	H	H	H	H

BCD 9's or 10's complement conversion

BCD words	Inputs E	D	C	B	A	Enable	Outputs Y8	Y7	Y6
0	L	L	L	L	L	L	H	L	H
1	L	L	L	L	H	"	H	L	L
2	L	L	L	H	L	"	L	H	H
3	L	L	L	H	H	"	L	H	L
4	L	L	H	L	L	"	L	H	H
5	L	L	H	L	H	"	L	H	L
6	L	L	H	H	L	"	L	L	H
7	L	L	H	H	H	"	L	L	L
8	L	H	L	L	L	"	L	L	H
9	L	H	L	L	H	"	L	L	L
0	H	L	L	L	L	"	L	L	L
1	H	L	L	L	H	"	H	L	L
2	H	L	L	H	L	"	H	L	L
3	H	L	L	H	H	"	L	H	H
4	H	L	H	L	L	"	L	H	H
5	H	L	H	L	H	"	L	H	L
6	H	L	H	H	L	"	L	H	L
7	H	L	H	H	H	"	L	L	H
8	H	H	L	L	L	"	L	L	H
9	H	H	L	L	H	"	L	L	L
Any	X	X	X	X	X	H	H	H	H

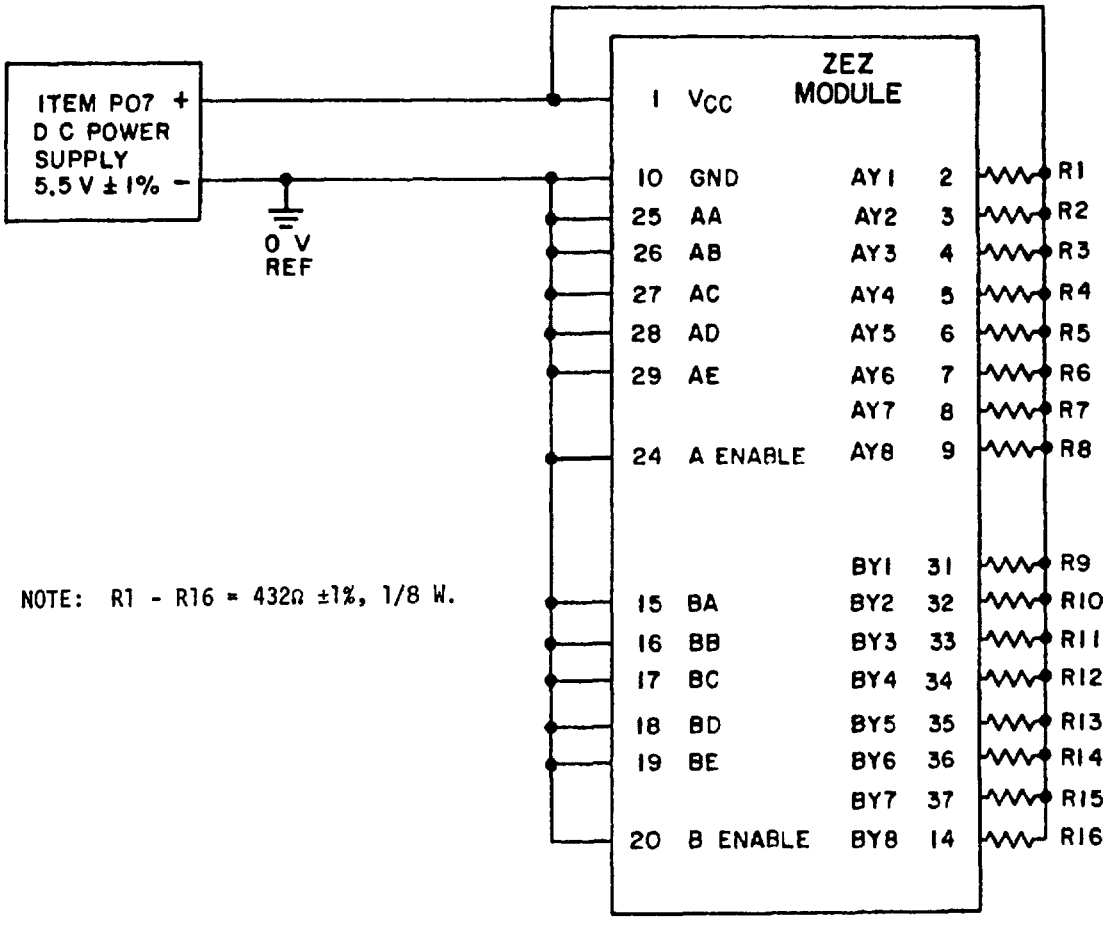
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NOTES:
1. R1 - R16 = 432Ω ±1%, 1/8 W.
2. R17 = 51.1Ω ±5%, 1/2 W.
3. Input pulse parameters:
   PRP = 100 ±100 ns.
   PWA = 500 ±50 ns.
   $t_{THL} = t_{TLH}$ = 10 ±2 ns.
   PA = 3 V.

FIGURE 3. Operating temperature and life test circuit.

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NOTE: R1 - R16 = 432Ω ±1%, 1/8 W.

* FIGURE 4. Component temperature test circuit.

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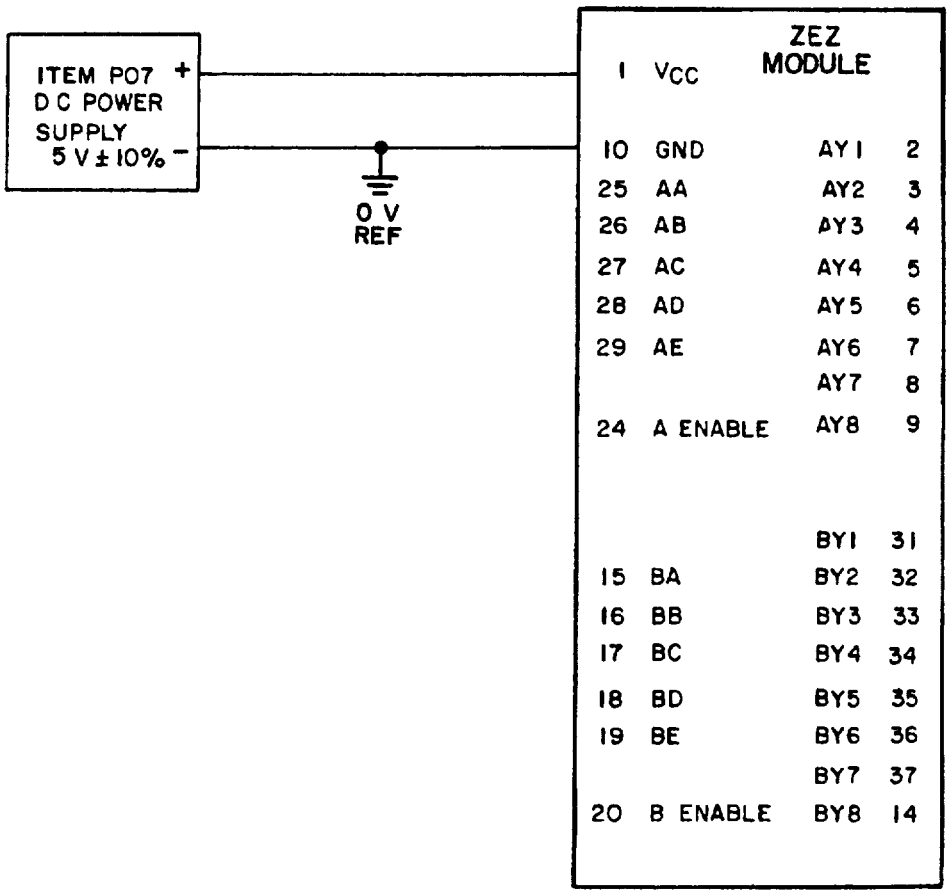


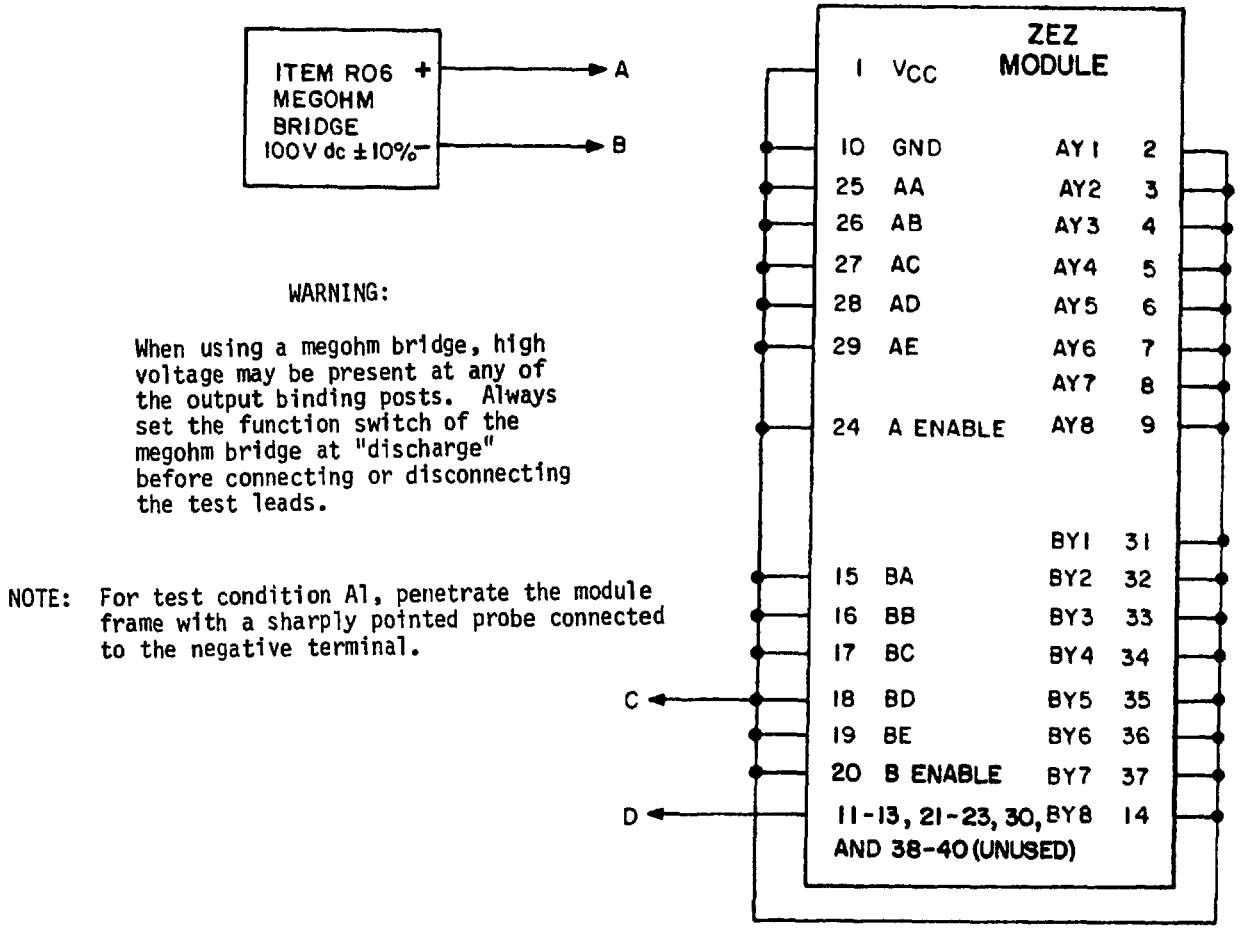
FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, D	Module frame	10	∞	MΩ	6
A2	Unused pin isolation	C	D	10	∞	MΩ	6

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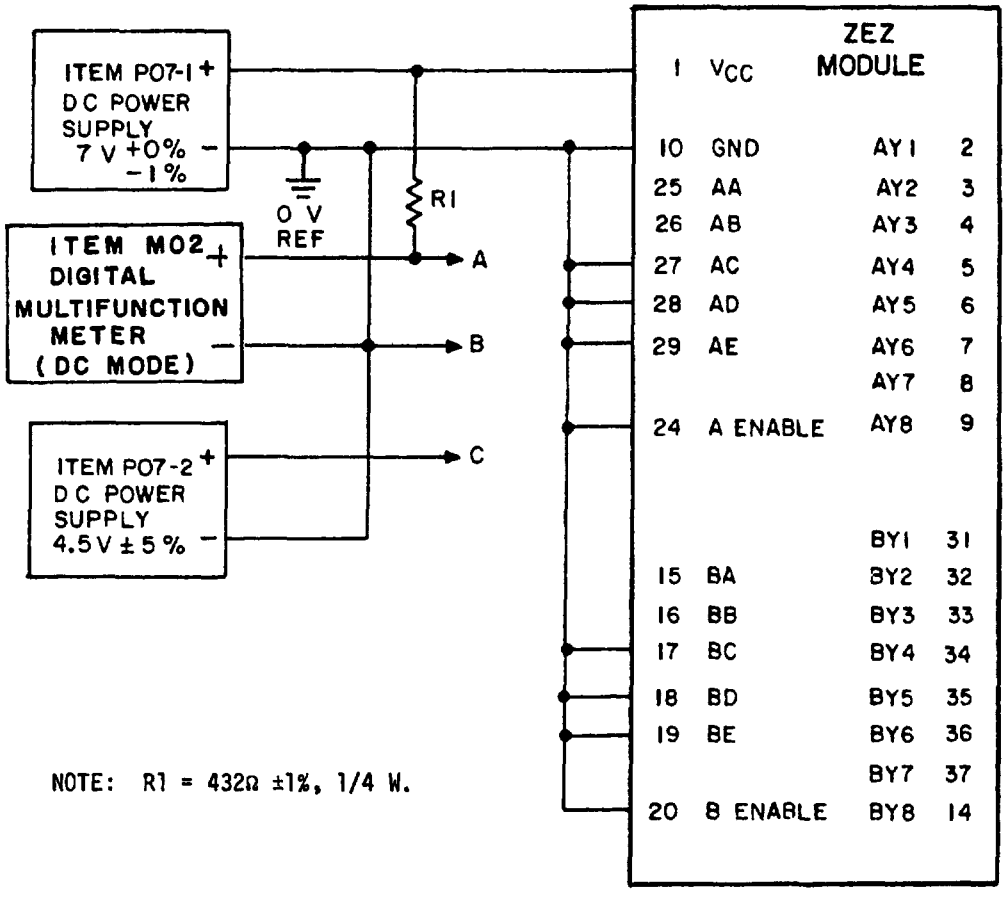
* FIGURE 6. Isolation test circuit for table III.

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TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
B1	AY1 V_{OUT} (low loaded)	2	25,26,15,16	Open	0	1	V	7
B2	AY2 "	3	"	"	"	"	"	"
B3	AY3 "	4	"	"	"	"	"	"
B4	AY4 "	5	"	"	"	"	"	"
B5	AY5 "	6	"	"	"	"	"	"
B6	AY7 "	8	"	"	"	"	"	"
B7	BY1 "	31	"	"	"	"	"	"
B8	BY2 "	32	"	"	"	"	"	"
B9	BY3 "	33	"	"	"	"	"	"
B10	BY4 "	34	"	"	"	"	"	"
B11	BY5 "	35	"	"	"	"	"	"
B12	BY7 "	37	"	"	"	"	"	"
B13	AY6 "	7	Open	25,26,15,16	"	"	"	"
B14	AY8 "	9	"	"	"	"	"	"
B15	BY6 "	36	"	"	"	"	"	"
B16	BY8 "	14	"	"	"	"	"	"

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NOTE: R1 = 432Ω ±1%, 1/4 W.

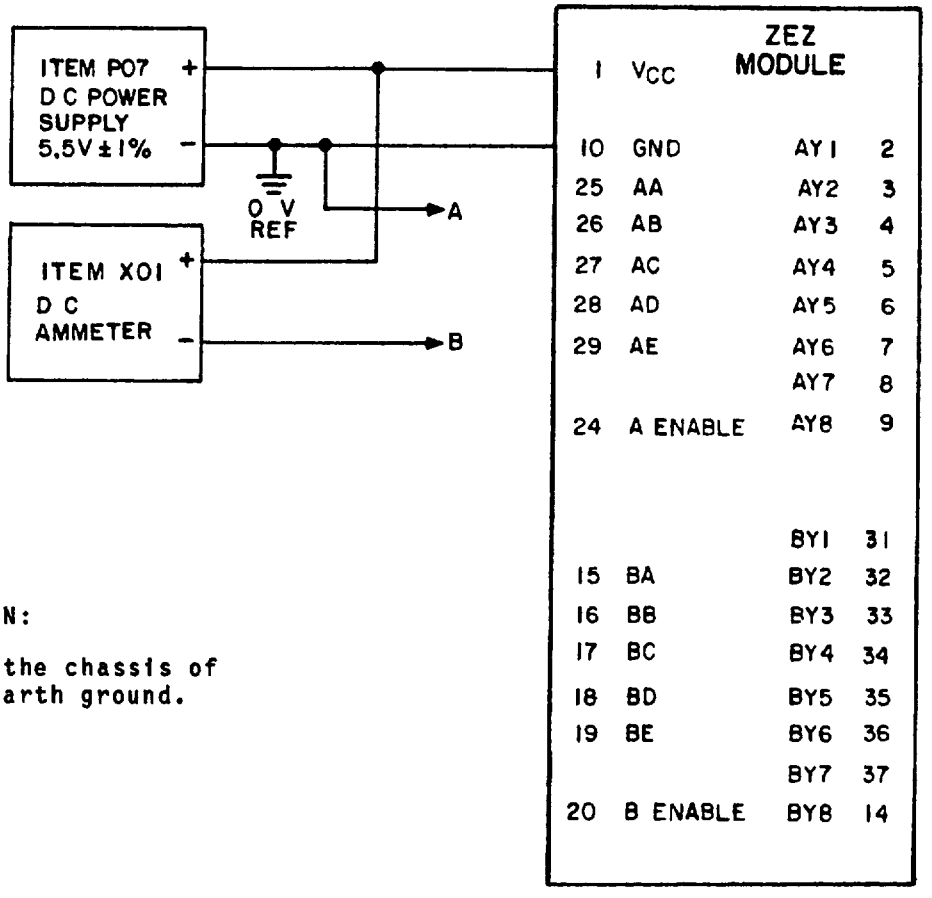
* FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	AA	Input breakdown current	26,27,28,29,24,15,16, 17,18,19,20	25	0	1	mA	8
C2	AB	"	25,27,28,29,24,15,16, 17,18,19,20	26	"	"	"	"
C3	AC	"	25,26,28,29,24,15,16, 17,18,19,20	27	"	"	"	"
C4	AD	"	25,26,27,29,24,15,16, 17,18,19,20	28	"	"	"	"
C5	AE	"	25,26,27,28,24,15,16, 17,18,19,20	29	"	"	"	"
C6	A Enable	"	25,26,27,28,29,15,16, 17,18,19,20	24	"	"	"	"
C7	BA	"	25,26,27,28,29,24,16, 17,18,19,20	15	"	"	"	"
C8	BB	"	25,26,27,28,29,24,15, 17,18,19,20	16	"	"	"	"
C9	BC	"	25,26,27,28,29,24,15, 16,18,19,20	17	"	"	"	"
C10	BD	"	25,26,27,28,29,24,15, 16,17,19,20	18	"	"	"	"
C11	BE	"	25,26,27,28,29,24,15, 16,17,18,20	19	"	"	"	"
C12	B	"	25,26,27,28,29,24,15, 16,17,18,19	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

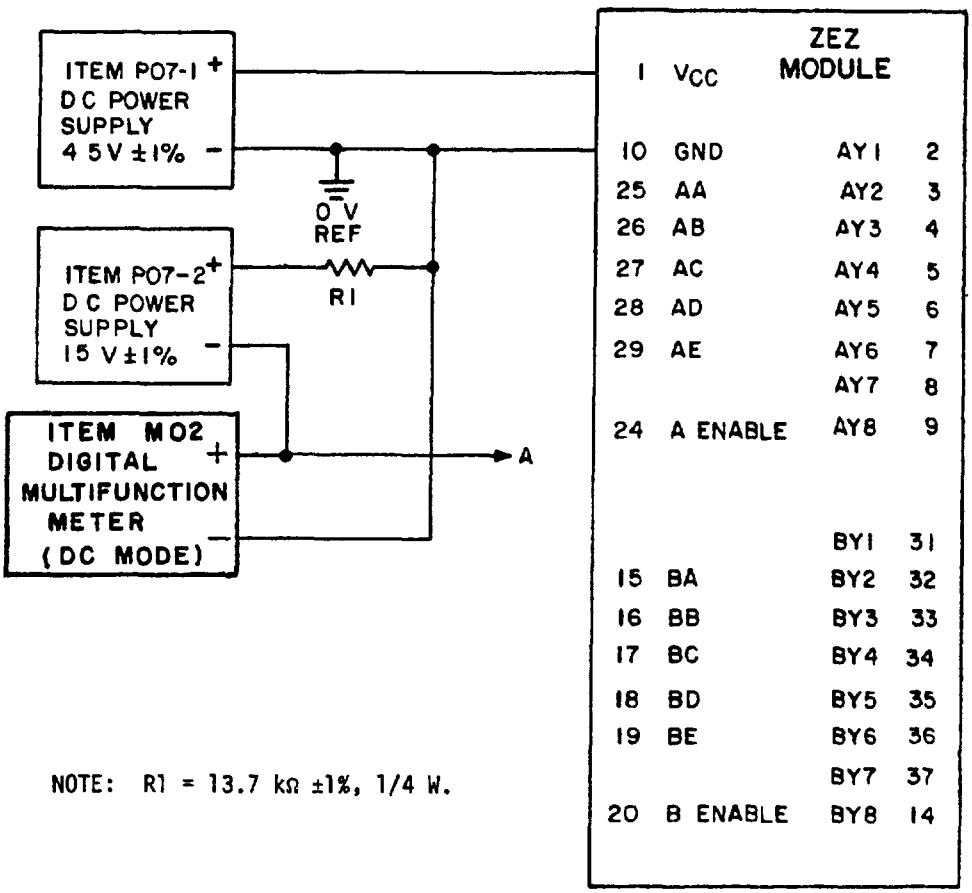
FIGURE 8. Input breakdown current test circuit for table V.

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*TABLE VI. Input clamp diode voltage test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	AA	Input clamp diode voltage	25	-0.2	-1.5	V	9
D2	AB	"	26	"	"	"	"
D3	AC	"	27	"	"	"	"
D4	AD	"	28	"	"	"	"
D5	AE	"	29	"	"	"	"
D6	A Enable	"	24	"	"	"	"
D7	BA	"	15	"	"	"	"
D8	BB	"	16	"	"	"	"
D9	BC	"	17	"	"	"	"
D10	BD	"	18	"	"	"	"
D11	BE	"	19	"	"	"	"
D12	B Enable	"	20	"	"	"	"

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NOTE: R1 = 13.7 kΩ ±1%, 1/4 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VI.

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Check the source to verify that this is the current version before use.

TABLE VII. Functional Test.

Test condition	Parameter	Connect		Output Pin																Test figure
		A to pin	B to pin	2	3	4	5	6	7	8	9	31	32	33	34	35	36	37	14	
E1	Functional	24,20,25,26,27,28, 29,15,16,17,18,19	Open	L	L	L	L	L	H	L	H	L	L	L	L	L	H	L	H	10
E2	"	24,20,26,27,28,29, 16,17,18,19	25,15	H	L	L	L	L	L	L	H	H	L	L	L	L	L	L	H	"
E3	"	24,20,25,27,28,29, 15,17,18,19	26,16	L	H	L	L	L	H	H	L	L	H	L	L	L	H	H	L	"
E4	"	24,20,27,28,29, 17,18,19	25,26, 15,16	H	H	L	L	L	L	H	L	H	H	L	L	L	L	H	L	"
E5	"	24,20,25,26,28,29, 15,16,18,19	27,17	L	L	H	L	L	H	H	L	L	L	H	L	L	H	H	L	"
E6	"	24,20,26,28,29, 16,18,19	25,27, 15,17	H	H	H	H	H	L	H	L	H	H	H	H	H	L	H	L	"
E7	"	24,20,25,28,29, 15,18,19	26,27, 16,17	H	H	H	H	H	H	L	L	H	H	H	H	H	H	L	L	"
E8	"	24,20,28,29,18,19	25,26, 27,15, 16,17	H	H	H	H	H	L	L	L	H	H	H	H	H	L	L	L	"
E9	"	24,20,25,26,27,29, 15,16,17,19	28,18,	H	L	H	L	L	H	L	L	H	L	H	L	L	H	L	L	"
E10	"	24,20,26,27,29, 16,17,19	25,28, 15,18	L	H	H	L	L	L	L	L	L	H	H	L	L	L	L	L	"
E11	"	24,20,25,27,29, 15,17,19	26,28, 16,18	H	H	H	L	L	H	H	H	H	H	H	L	L	H	H	H	"
E12	"	24,20,27,29,17,19	25,26, 28,15 16,18	L	L	L	H	L	H	H	H	L	L	L	H	L	H	H	H	"
E13	"	24,20,25,26,29, 15,16,19	27,28, 17,18	H	L	L	H	L	H	H	H	H	L	L	H	L	H	H	H	"
E14	"	24,20,26,29,16,19	25,27, 28,15 17,18	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

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TABLE VII. Functional Test - Continued.

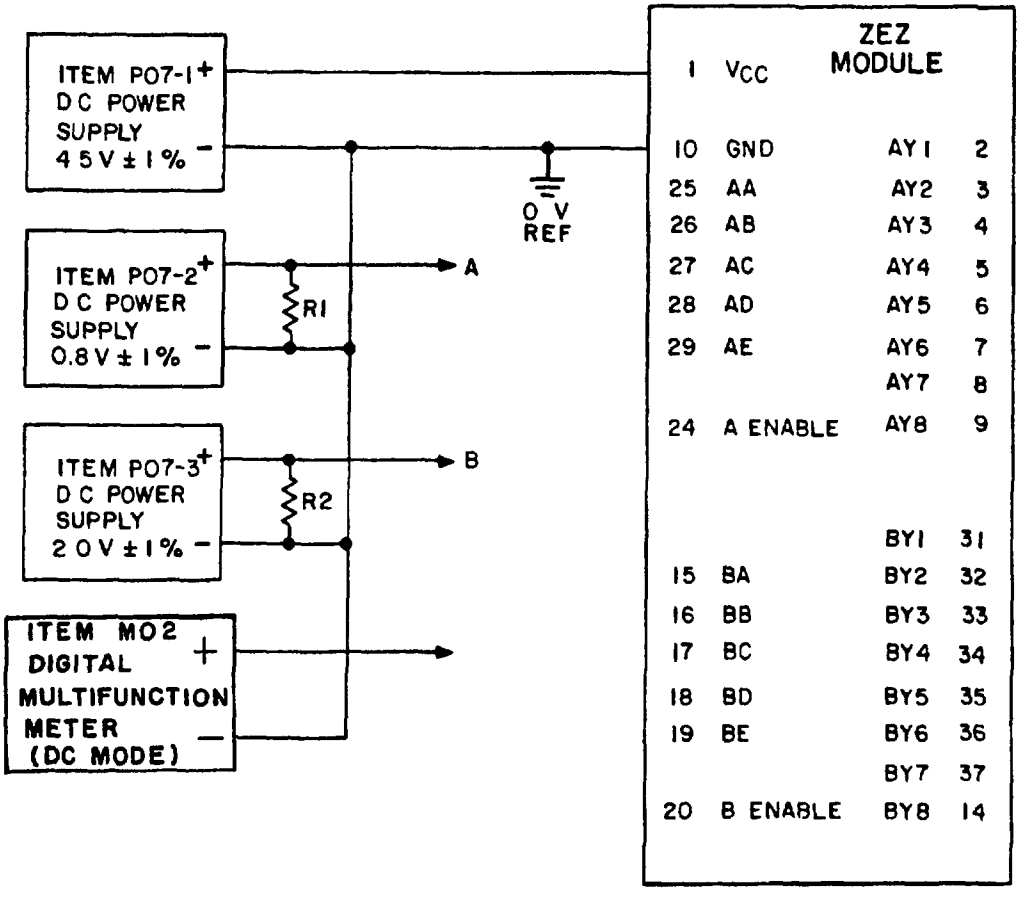
Test condition	Parameter	Connect		Output Pin															Test figure
		A to pin	B to pin	2	3	4	5	6	7	8	9	31	32	33	34	35	36	37	14	
E15	Functional	24,20,25,29,15,19	26,27, 28,15, 17,18	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	10
E16	"	24,20,29,19	25,26, 27,28, 15,16, 17,18	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E17	"	24,20,25,26,27,28, 15,16,17,18	29,19	L	H	L	H	L	L	L	L	L	H	L	H	L	L	L	L	"
E18	"	24,20,26,27,28, 16,17,18	25,29, 15,19	H	H	L	H	L	L	L	H	H	H	L	H	L	L	L	H	"
E19	"	24,20,25,27,28, 15,17,18	26,29, 16,19	L	L	H	H	L	L	L	H	L	L	H	H	L	L	L	H	"
E20	"	24,20,27,28,17,18	25,26, 29,15, 16,19	H	L	H	H	L	H	H	L	H	L	H	H	L	H	H	L	"
E21	"	24,20,25,26,28,15, 16,18	27,29, 17,19	L	H	H	H	L	H	H	L	L	H	H	H	L	H	H	L	"
E22	"	24,20,26,28,16,18	25,27, 29,15, 17,19	H	H	H	H	H	L	H	L	H	H	H	H	H	L	H	L	"
E23	"	24,20,25,28,15,18	26,27, 29,16, 17,19	H	H	H	H	H	L	H	L	H	H	H	H	H	L	H	L	"
E24	"	24,20,28,18	25,26, 27,29, 15,16, 17,19	H	H	H	H	H	H	L	L	H	H	H	H	H	H	L	L	"
E25	"	24,20,25,26,27, 15,16,17	28,29, 18,19	H	H	H	H	L	H	L	L	H	H	H	H	L	H	L	L	"
E26	"	24,20,26,27,16,17	25,28, 29,15, 18,19	L	L	L	L	H	L	L	L	L	L	L	L	H	L	L	L	"

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TABLE VII. Functional Test - Continued.

Test condition	Parameter	Connect		Output Pin															Test figure
		A to pin	B to pin	2	3	4	5	6	7	8	9	31	32	33	34	35	36	37	14	
E27	Functional	24,20,25,27,15,17	26,28, 29,16, 18,19	H	L	L	L	H	H	H	H	H	L	L	L	H	H	H	H	10
E28	"	24,20,27,17	25,26, 28,29, 15,16, 18,19	L	H	L	L	H	H	H	H	L	H	L	L	H	H	H	H	"
E29	"	24,20,25,26,15,16	27,28, 29,17, 18,19	H	H	L	L	H	H	H	H	H	H	L	L	H	H	H	H	"
E30	"	24,20,26,16	25,27, 28,29, 15,17, 18,19	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E31	"	24,20,25,15	26,27, 28,29, 16,17, 18,19	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E32	"	24,20	25,26, 27,28, 29,15, 16,17 18,19	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E33	"	25,26,27,28,29,15, 16,17,18,19	24,20	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"
E34	"	25,26,27,28,15, 16,17,18	24,20 29,19	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. Use item M02 to measure output logic level.
3. Load outputs as shown on figure 18.

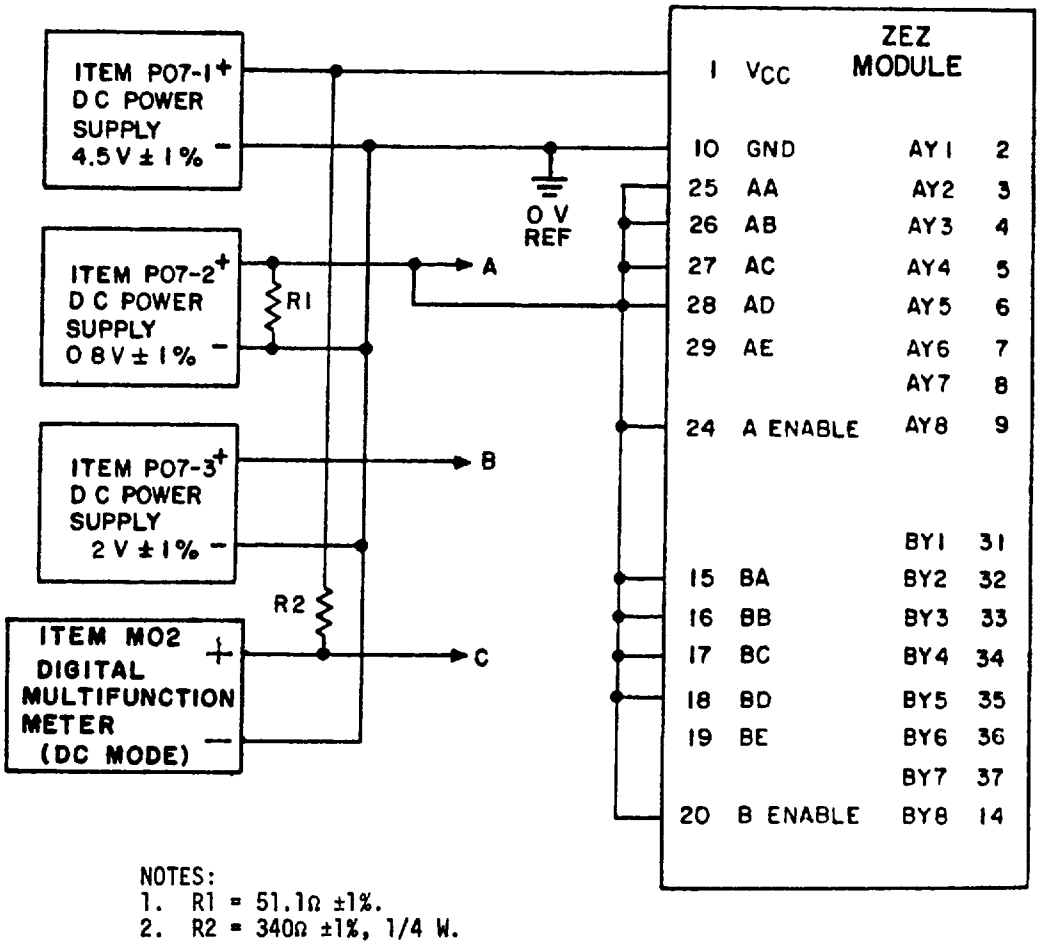
* FIGURE 10. Functional test circuit for table VII.

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*TABLE VIII. Output voltage (low-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
F1	AY1	Output voltage, low level	19,29	Open	2	0	400	mV	11
F2	AY2	"	"	"	3	"	"	"	"
F3	AY3	"	"	"	4	"	"	"	"
F4	AY4	"	"	"	5	"	"	"	"
F5	AY5	"	"	"	6	"	"	"	"
F6	AY7	"	"	"	8	"	"	"	"
F7	BY1	"	"	"	31	"	"	"	"
F8	BY2	"	"	"	32	"	"	"	"
F9	BY3	"	"	"	33	"	"	"	"
F10	BY4	"	"	"	34	"	"	"	"
F11	BY5	"	"	"	35	"	"	"	"
F12	BY7	"	"	"	37	"	"	"	"
F13	AY6	"	Open	19,29	7	"	"	"	"
F14	AY8	"	"	"	9	"	"	"	"
F15	BY6	"	"	"	36	"	"	"	"
F16	BY8	"	"	"	14	"	"	"	"

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NOTES:
1. R1 = 51.1Ω ±1%.
2. R2 = 340Ω ±1%, 1/4 W.

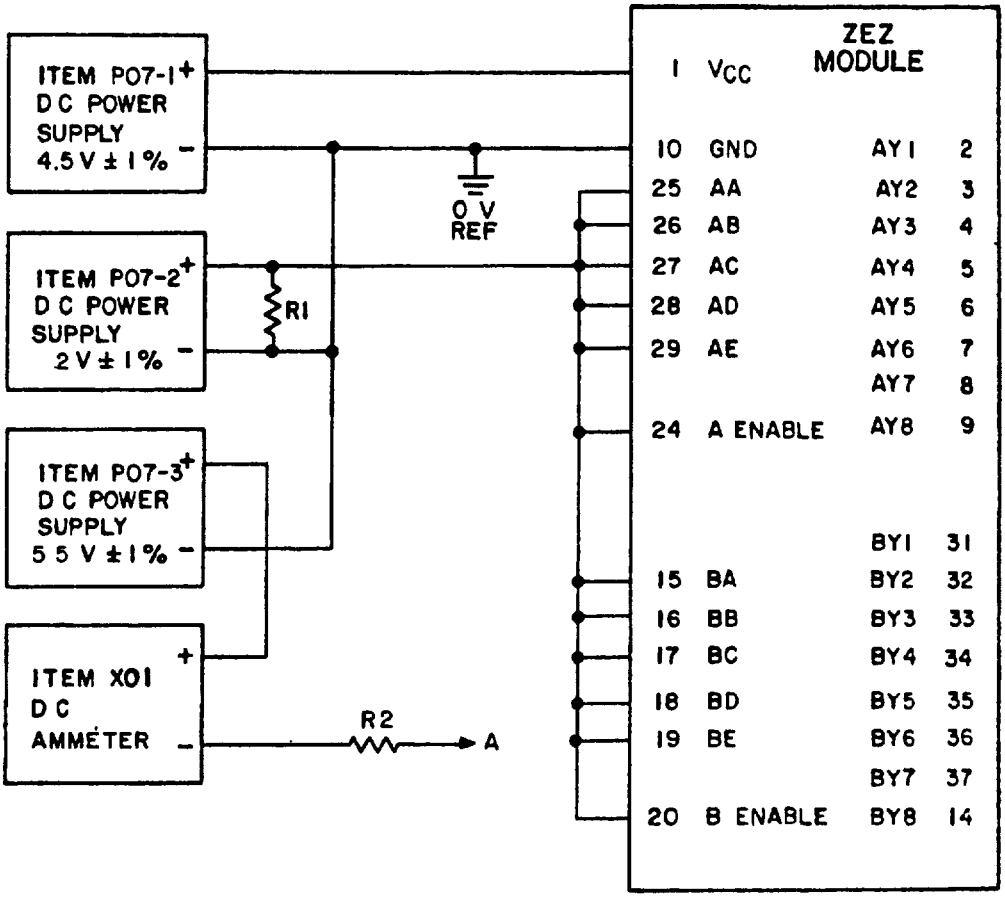
* FIGURE 11. Output voltage (low-level) test circuit for table VIII.

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TABLE IX. Output leakage current (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
G1	AY1 Output leakage current, high level	2	0	100	µA	12
G2	AY2 "	3	"	"	"	"
G3	AY3 "	4	"	"	"	"
G4	AY4 "	5	"	"	"	"
G5	AY5 "	6	"	"	"	"
G6	AY6 "	7	"	"	"	"
G7	AY7 "	8	"	"	"	"
G8	AY8 "	9	"	"	"	"
G9	BY1 "	31	"	"	"	"
G10	BY2 "	32	"	"	"	"
G11	BY3 "	33	"	"	"	"
G12	BY4 "	34	"	"	"	"
G13	BY5 "	35	"	"	"	"
G14	BY6 "	36	"	"	"	"
G15	BY7 "	37	"	"	"	"
G16	BY8 "	14	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 422Ω ±1%, 1/8 W.

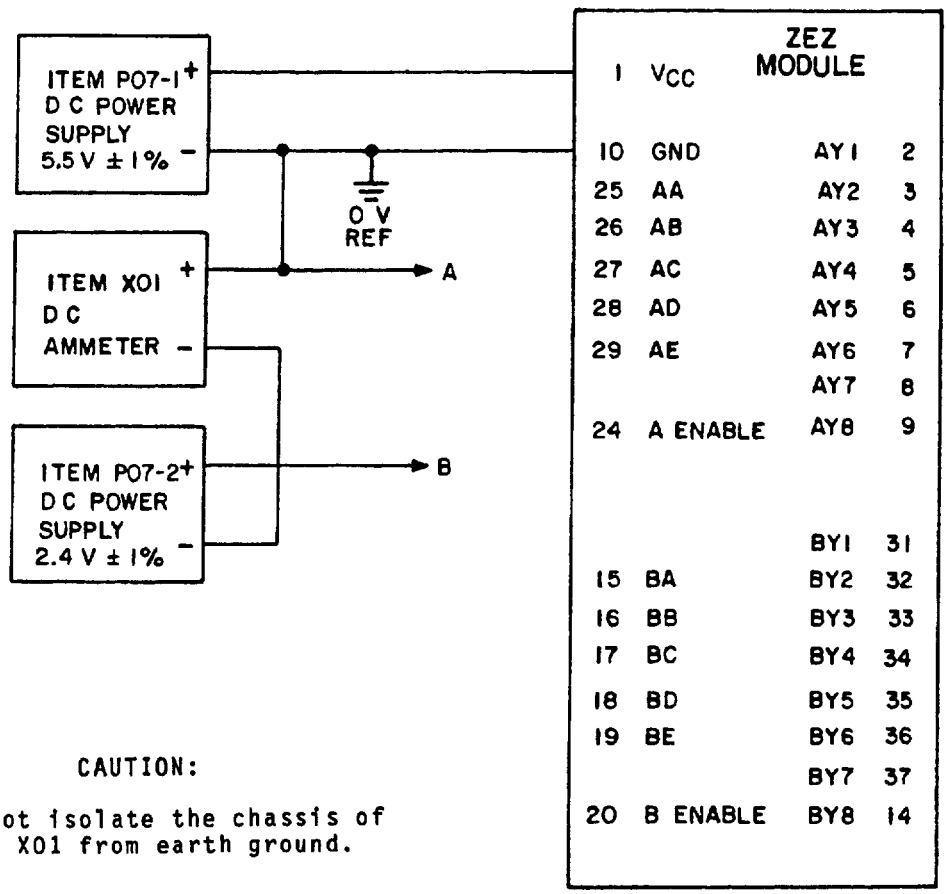
FIGURE 12. Output leakage current (high-level) test circuit for table IX.

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TABLE X. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H1	AA	Input current, high level	26,27,28,29,24,15,16, 17,18,19,20	25	0	40	µA	13
H2	AB	"	25,27,28,29,24,15,16, 17,18,19,20	26	"	"	"	"
H3	AC	"	25,26,28,29,24,15,16, 17,18,19,20	27	"	"	"	"
H4	AD	"	25,26,27,29,24,15,16, 17,18,19,20	28	"	"	"	"
H5	AE	"	25,26,27,28,24,15,16, 17,18,19,20	29	"	"	"	"
H6	A Enable	"	25,26,27,28,29,15,16, 17,18,19,20	24	"	"	"	"
H7	BA	"	25,26,27,28,29,24,16, 17,18,19,20	15	"	"	"	"
H8	BB	"	25,26,27,28,29,24,15, 17,18,19,20	16	"	"	"	"
H9	BC	"	25,26,27,28,29,24,15, 16,18,19,20	17	"	"	"	"
H10	BD	"	25,26,27,28,29,24,15, 16,17,19,20	18	"	"	"	"
H11	BE	"	25,26,27,28,29,24,15, 16,17,18,20	19	"	"	"	"
H12	B Enable	"	25,26,27,28,29,24,15, 16,17,18,19	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

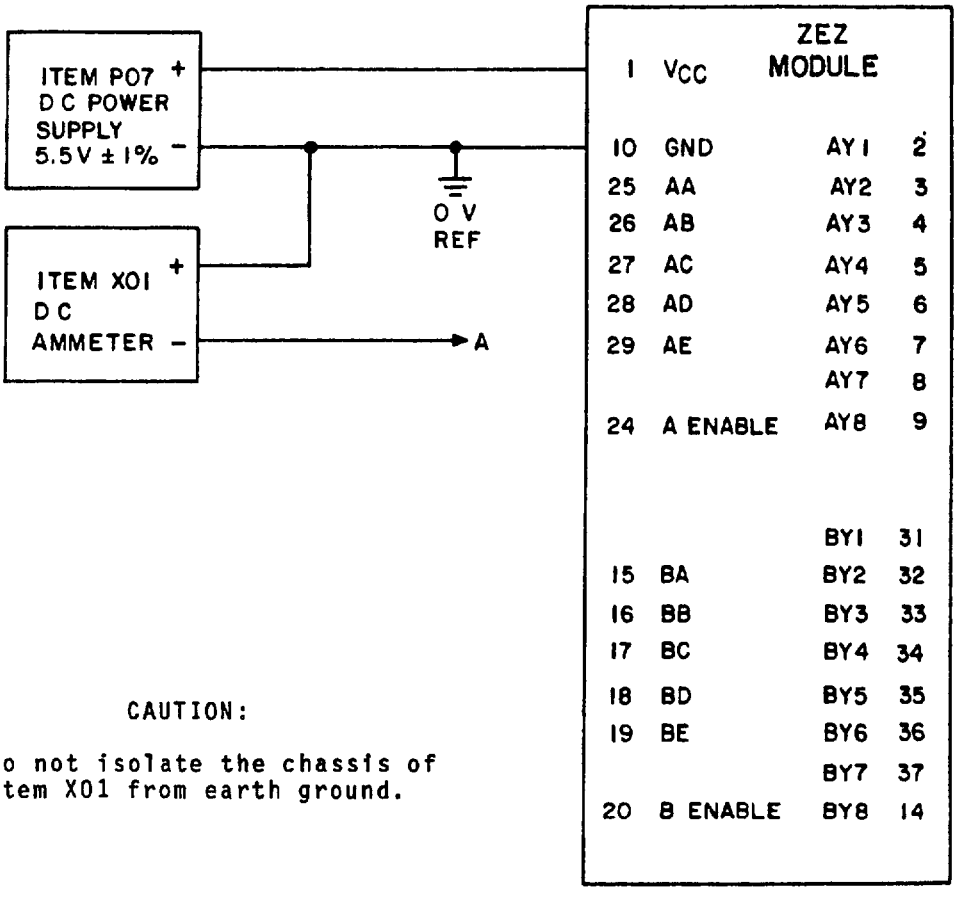
FIGURE 13. Input current (high-level) test circuit for table X.

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*TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	AA Input current, low level	25	0.00	-1.08	mA	14
J2	AB "	26	"	"	"	"
J3	AC "	27	"	"	"	"
J4	AD "	28	"	"	"	"
J5	AE "	29	"	"	"	"
J6	A Enable "	24	"	"	"	"
J7	BA "	15	"	"	"	"
J8	BB "	16	"	"	"	"
J9	BC "	17	"	"	"	"
J10	BD "	18	"	"	"	"
J11	BE "	19	"	"	"	"
J12	B Enable "	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 14. Input current (low-level) test circuit for table XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K	Power supply current	2	198	mA	15

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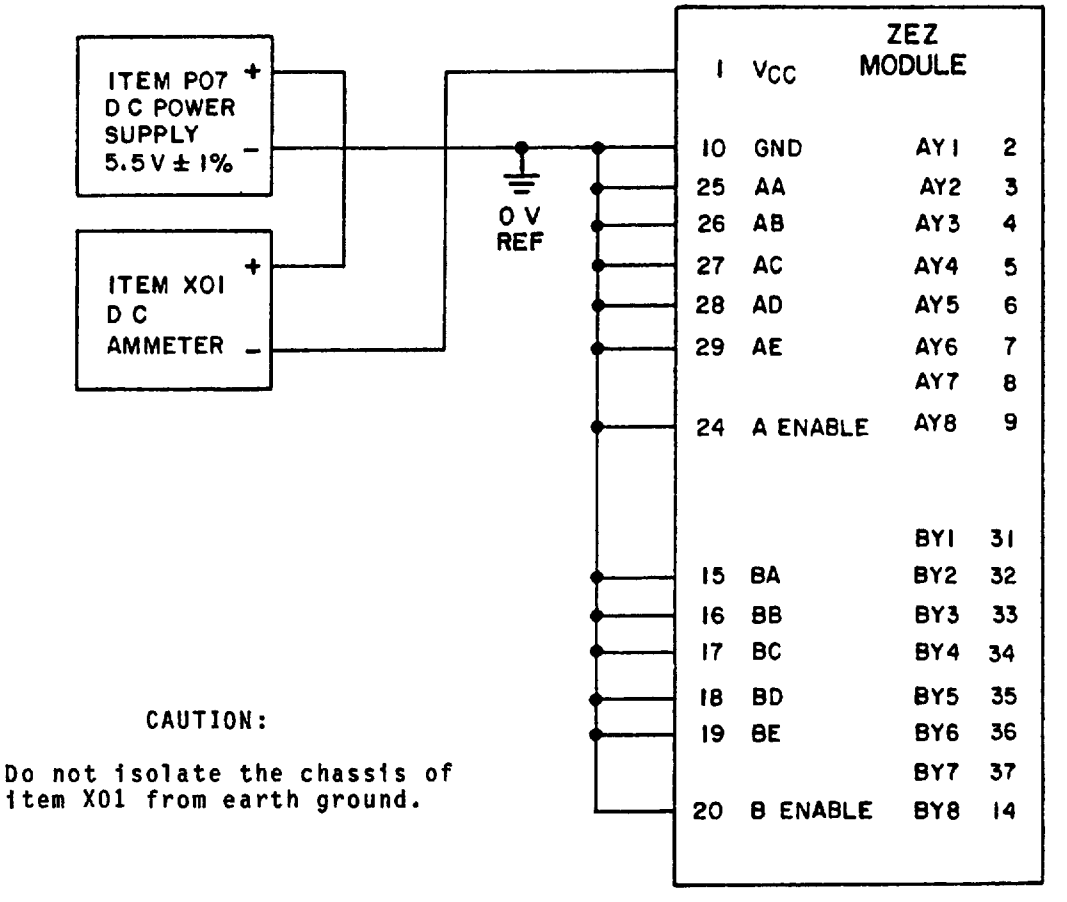


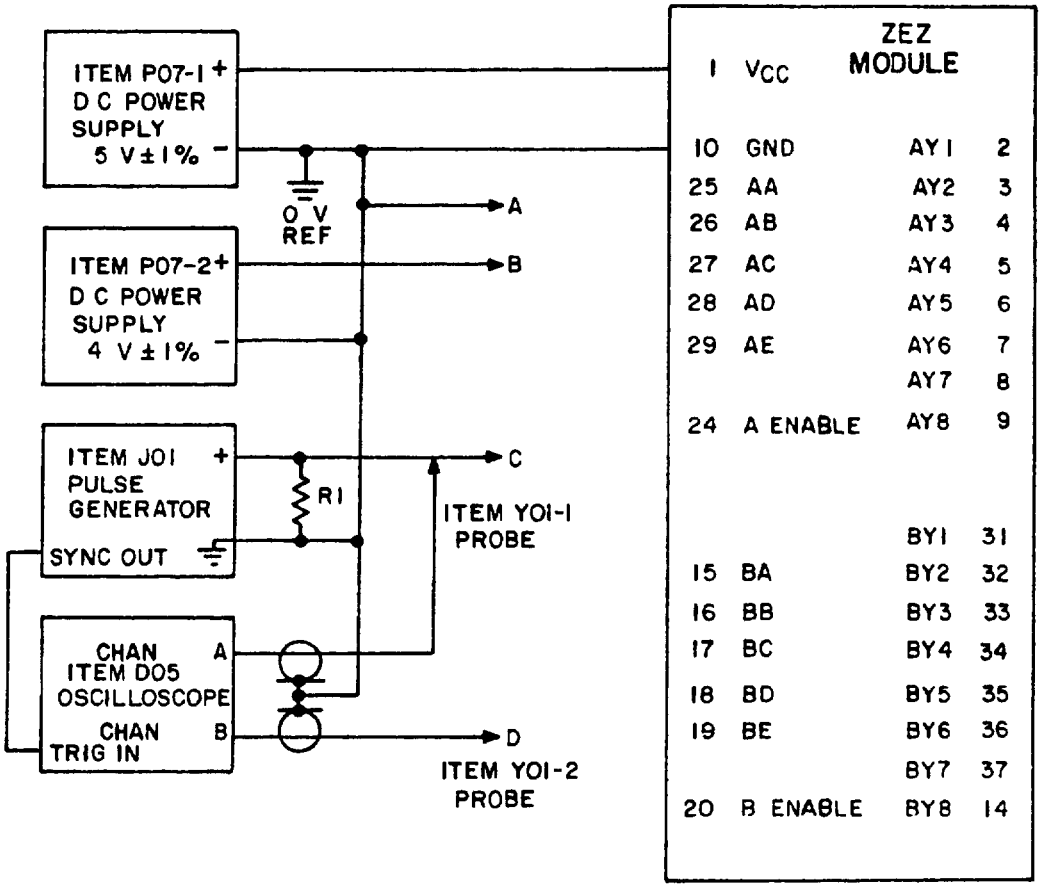
FIGURE 15. Power supply current test circuit for table XII.

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TABLE XIII. Transition time (t_{THL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	AY1 Transition time (t_{THL})	25,26,27, 28,29	Open	24	2	5	30	ns	16,17
L2	AY2 "	"	"	"	3	"	"	"	"
L3	AY3 "	"	"	"	4	"	"	"	"
L4	AY4 "	"	"	"	5	"	"	"	"
L5	AY5 "	"	"	"	6	"	"	"	"
L6	AY6 "	26,27,28	25,29	"	7	"	"	"	"
L7	AY7 "	25,26,27, 28,29	Open	"	8	"	"	"	"
L8	AY8 "	27,28	25,26,29	"	9	"	"	"	"
L9	BY1 "	15,16,17, 18,19	Open	20	31	"	"	"	"
L10	BY2 "	"	"	"	32	"	"	"	"
L11	BY3 "	"	"	"	33	"	"	"	"
L12	BY4 "	"	"	"	34	"	"	"	"
L13	BY5 "	"	"	"	35	"	"	"	"
L14	BY6 "	16,17,18	15,19	"	36	"	"	"	"
L15	BY7 "	15,16,17, 18,19	Open	"	37	"	"	"	"
L16	BY8 "	17,18	15,16,19	"	14	"	"	"	"

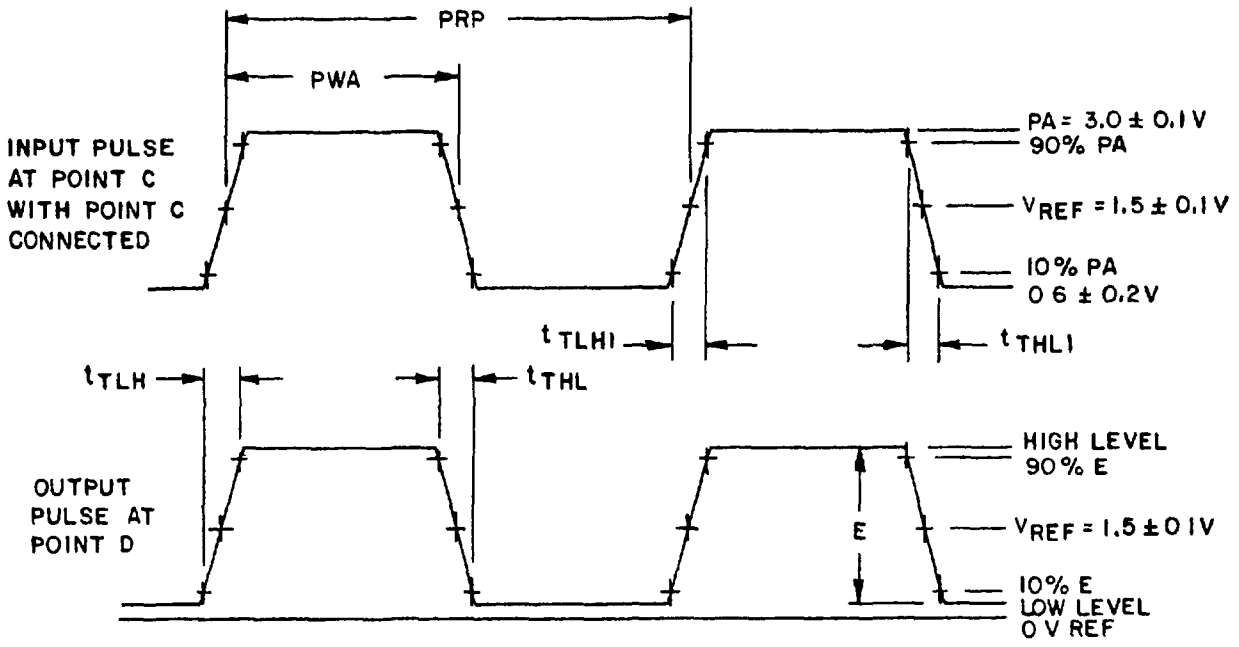
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load output as shown on figure 18.

FIGURE 16. Transition time (t_{THL}) test circuit for table XIII.

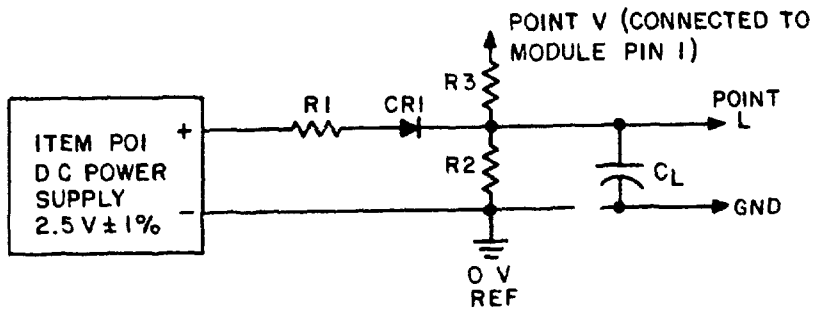
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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PRP = 1,500 ±50 ns.
   PWA = 750 ±50 ns.
2. Before making the measurements verify that the output pulse high level $\geq$ 2.4 V and that the low level $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 17. Transition time (t_{THL}) test circuit waveforms for table XIII.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 12 mA ±5% out of point L with point L held at 0.4 V and point V held at 5 V.
3. R2 = 9.09 kΩ ±1%, 1/8 W.
4. R3 = 2.87 kΩ ±1%, 1/8 W.
5. C_L = 50 ±5 pF (including probe and parasitic capacitance).
6. Lead length from load to module output pin shall be less than 2 inches (50.80 mm).

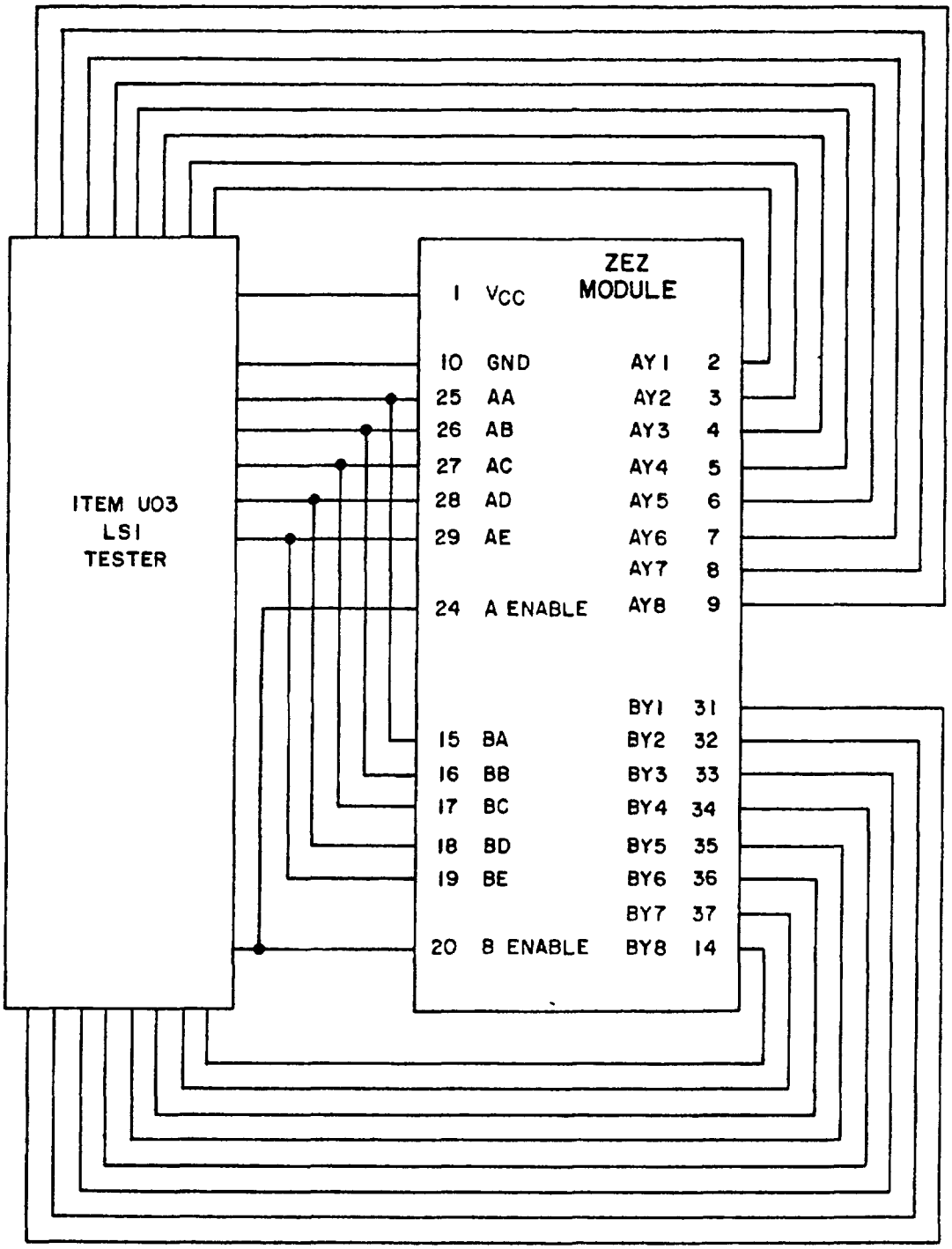
FIGURE 18. ZEZ module load circuit.

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TABLE XIV. Galloping read access pattern test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
M1	Galloping read access pattern	Module shall conform to the requirements of figures 19 and 20 for each test condition.	No failure			19,20
M2	Galloping read access pattern		No failure			19,20

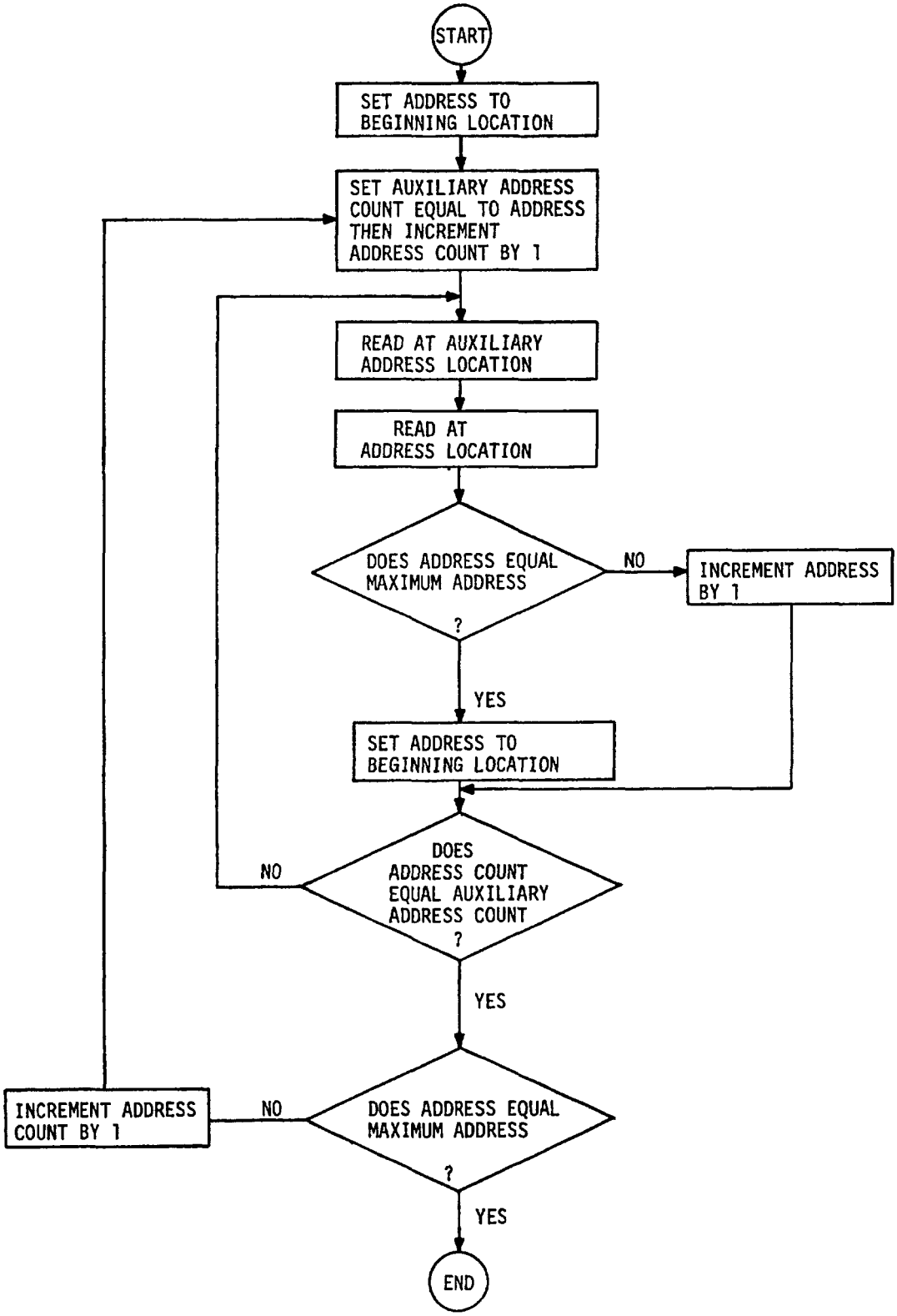
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NOTES:
1. V_{CC} = 4.5 V ±1% for test condition M1.
2. V_{CC} = 5.5 V ±1% for test condition M2.
3. Load outputs as shown on figure 18.

FIGURE 19. Galloping read access pattern test circuit for table XIV.

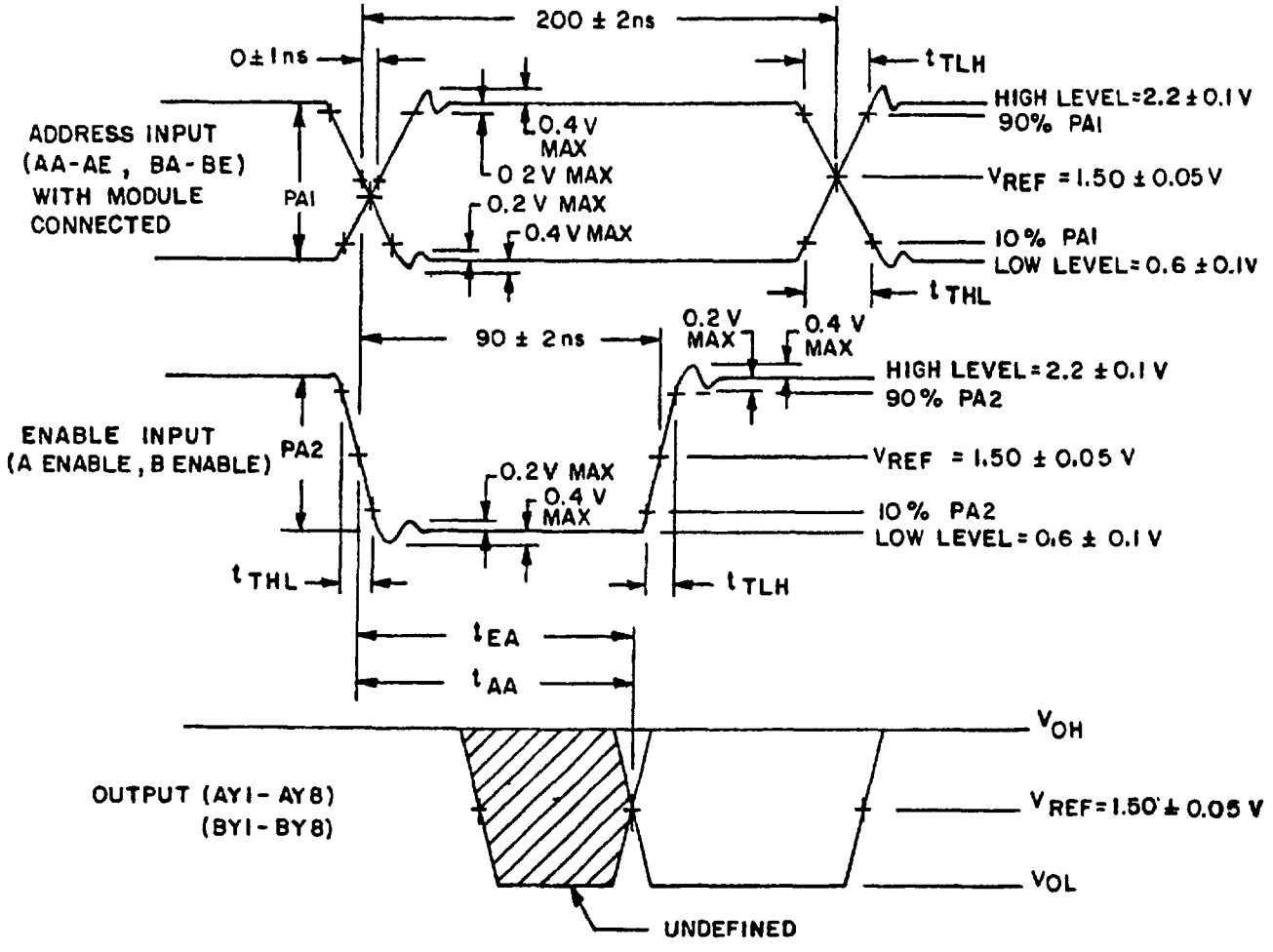
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NOTE: Pattern is read and verified at each location.

FIGURE 20. Galloping read access pattern program flow chart, waveforms, and program for table XIV.

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NOTES:
1. Address access time t_{AA} (50 pF) = 55 $^{+0}_{-4}$ ns.
2. Enable access time t_{EA} (50 pF) = 45 $^{+0}_{-4}$ ns.
3. $t_{TLH} = t_{THL}$ = 10 ±4 ns.

FIGURE 20. Galloping read access pattern program flow chart, waveforms, and program for table XIV - Continued.

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Program for test conditions M1-M2:

1. The objective of the galloping read access pattern is to test address access time. It is necessary to test all possible address transitions. The test also verifies circuit enable time (t_{EA}) and address access time (t_{AA}).

2. A program that fulfills the galloping read access test objective would be:

(a) Read memory address zero.
(b) Read memory address one.
(c) Read memory address zero.
(d) Read memory address two.
(e) Read memory address zero.
(f) Read memory address three.

•
•
•

(g) Read memory address zero.
(h) Read memory address 31.
(i) Read memory address one.
(j) Read memory address two.
(k) Read memory address one.
(l) Read memory address three.

•
•
•

(m) Read memory address one.
(n) Read memory address 31.
(o) Read memory address one.
(p) Read memory address zero.
(q) Read memory address two.
(r) Read memory address three.
(s) Read memory address two.
(t) Read memory address four.

•
•
•

(u) Continue this pattern until all memory addresses have been tested to all other memory addresses.

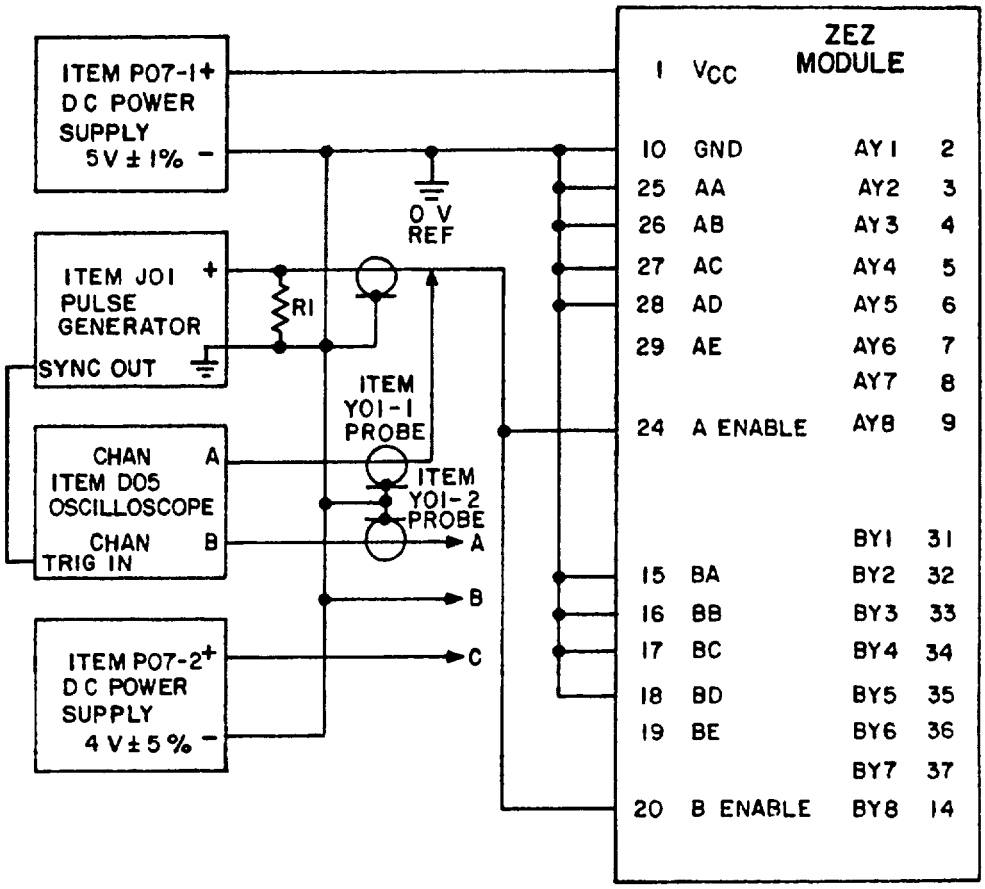
FIGURE 20. Galloping read access pattern program flow chart, waveforms, and program for table XIV - Continued.

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TABLE XV. Enable recovery time test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	AY1 Enable Recovery time	2	19,29	Open	5	55	ns	21,22
N2	AY2 "	3	"	"	"	"	"	"
N3	AY3 "	4	"	"	"	"	"	"
N4	AY4 "	5	"	"	"	"	"	"
N5	AY5 "	6	"	"	"	"	"	"
N6	AY7 "	8	"	"	"	"	"	"
N7	BY1 "	31	"	"	"	"	"	"
N8	BY2 "	32	"	"	"	"	"	"
N9	BY3 "	33	"	"	"	"	"	"
N10	BY4 "	34	"	"	"	"	"	"
N11	BY5 "	35	"	"	"	"	"	"
N12	BY7 "	37	"	"	"	"	"	"
N13	AY6 "	7	Open	19,29	"	"	"	"
N14	AY8 "	9	"	"	"	"	"	"
N15	BY6 "	36	"	"	"	"	"	"
N16	BY8 "	14	"	"	"	"	"	"

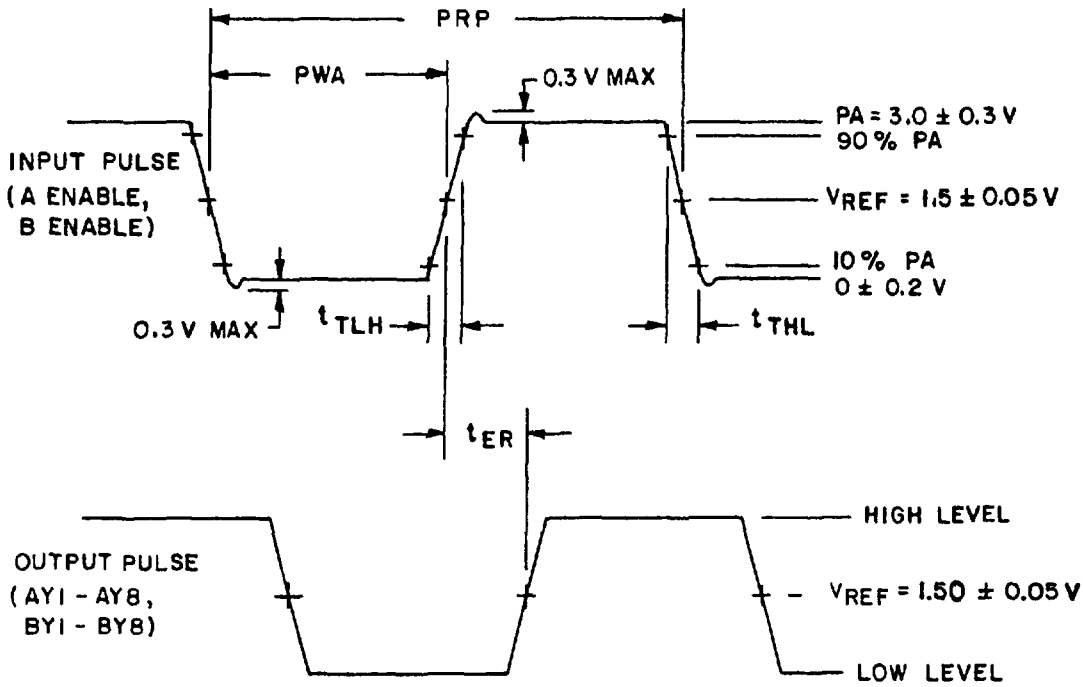
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. Load outputs as shown on figure 18.

FIGURE 21. Enable recovery time test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   PRP = 1000 ±100 ns.
   PWA = 500 ±50 ns.
   $t_{THL} = t_{TLH}$ = 10 ±2 ns.
2. Before making the measurements verify that the output pulse high level $\geq$ 2.4 V and that the low level $\leq$ 0.4 V.
3. All voltages are referenced to ground.

FIGURE 22. Enable recovery time test circuit waveforms for table XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Power supply transient amplitude	0	400	mV	23,24

TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q	Power supply filter capacitance	Open	1	3.2	6.2	µF	25

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TABLE XVIII. Input capacitance test.

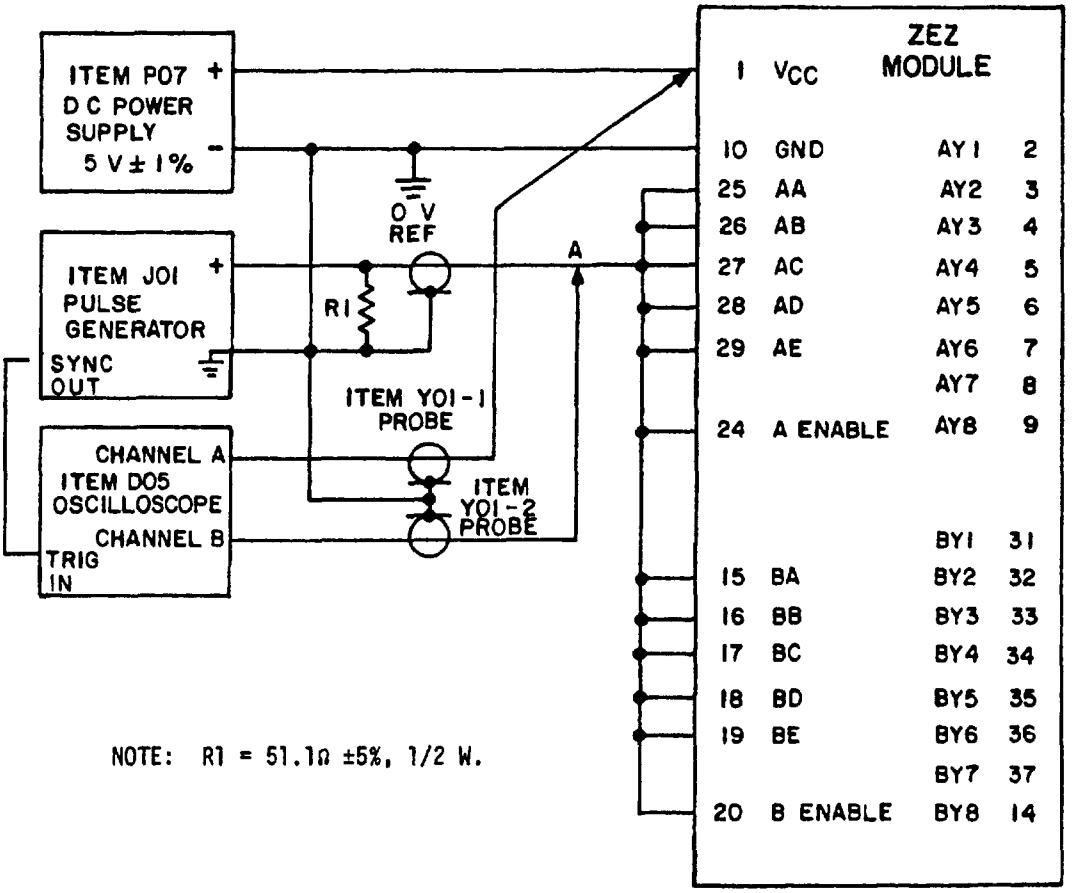
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
R1	AA	Input capacitance	25	Open	2	15	pF	25
R2	AB	"	26	"	"	"	"	"
R3	AC	"	27	"	"	"	"	"
R4	AD	"	28	"	"	"	"	"
R5	AE	"	29	"	"	"	"	"
R6	A Enable	"	24	"	"	"	"	"
R7	BA	"	15	"	"	"	"	"
R8	BB	"	16	"	"	"	"	"
R9	BC	"	17	"	"	"	"	"
R10	BD	"	18	"	"	"	"	"
R11	BE	"	19	"	"	"	"	"
R12	B Enable	"	20	"	"	"	"	"

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TABLE XIX. Output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
S1	AY1 Output capacitance	2	Open	2	15	pF	25
S2	AY2 "	3	"	"	"	"	"
S3	AY3 "	4	"	"	"	"	"
S4	AY4 "	5	"	"	"	"	"
S5	AY5 "	6	"	"	"	"	"
S6	AY6 "	7	"	"	"	"	"
S7	AY7 "	8	"	"	"	"	"
S8	AY8 "	9	"	"	"	"	"
S9	BY1 "	31	"	"	"	"	"
S10	BY2 "	32	"	"	"	"	"
S11	BY3 "	33	"	"	"	"	"
S12	BY4 "	34	"	"	"	"	"
S13	BY5 "	35	"	"	"	"	"
S14	BY6 "	36	"	"	"	"	"
S15	BY7 "	37	"	"	"	"	"
S16	BY8 "	14	"	"	"	"	"

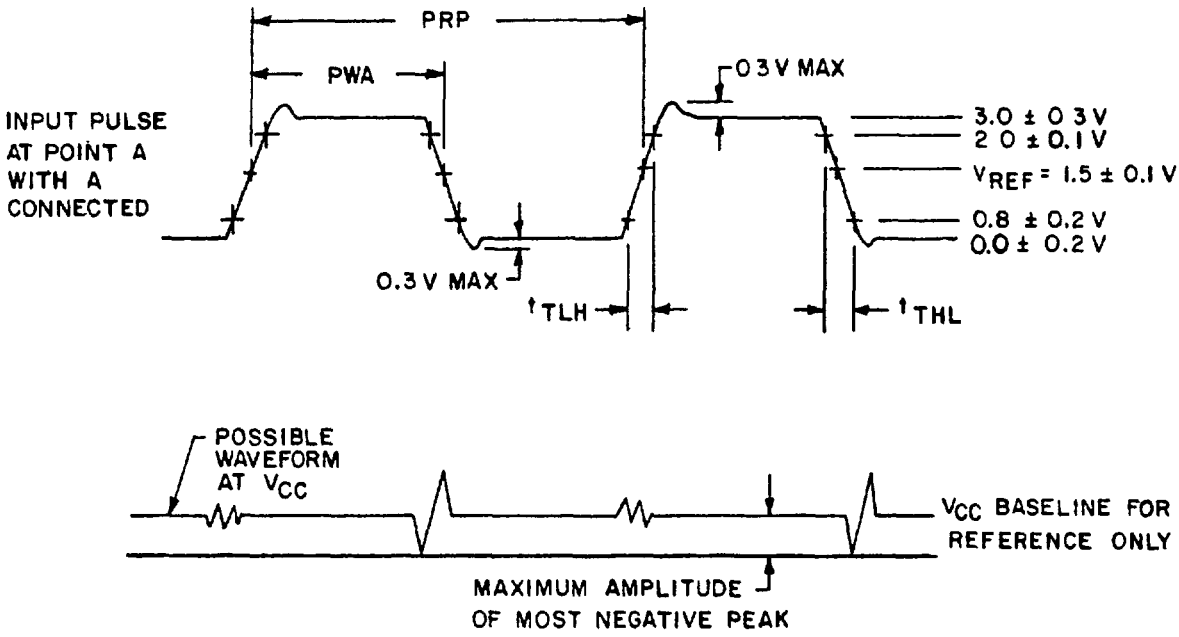
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NOTE: R1 = 51.1Ω ±5%, 1/2 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVI.

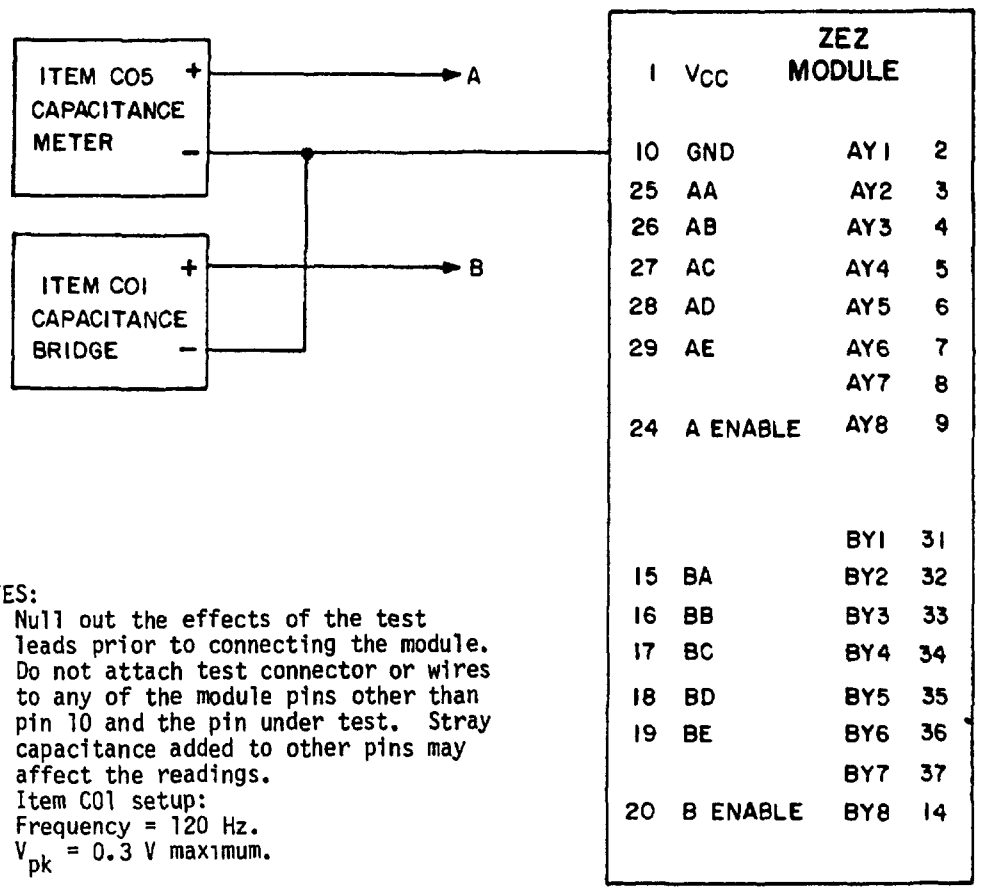
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NOTE: Input pulse parameters:
PWA = 1.0 ±0.1 µs.
PRP = 2.0 ±0.2 µs.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns verified at each input.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVI.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 25. Power supply filter capacitance, input, and output capacitance test circuit for tables XVII, XVIII and XIX.

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